

SIMA Research - Project Plan with Claude

Overview

Goal: Write a narrative review article on imputation for multivariate time series, focused on air pollution data.

Working title: *From Statistical Missing-Data Inference to Deep Generative Imputation for Multivariate Time Series: A Narrative Review*

Team: Valeria, Fernando, Arturo + 2 professors (including Facundo)

Context: College degree research project. The review will later inform the development of an imputation method for SIMA (Sistema Integral de Monitoreo Ambiental) in Monterrey.

Phase 1: Citation Graph Tool (DONE)

Objective: Build an interactive, directed citation graph of the ~75 articles in the Zotero collection. Use it to:

- Identify which articles are central (many connections)
- Spot isolated articles that may not belong
- Discover clusters/communities (classical, air quality, deep learning, etc.)
- Guide decisions about what to keep/discard

Approach:

- Language: Python
- Data source: Export from Zotero (.bib already in repo)
- Citation lookup: OpenAlex API (by DOI, with title fallback)
- Forward + reverse citation matching, canonical version resolution
- Visualization: Interactive pyvis — runs locally in browser
- Graph is directed: edge from A to B means "A cites B"

Input: SIMA_PERSONAL.bib

Output: Interactive HTML graph + summary of isolated/central nodes

Tasks

- ☒ Parse .bib file, extract metadata (DOI, title, authors, year)
 - ☒ Query OpenAlex for each article's references
 - ☒ Match references against other articles in the collection
 - ☒ Build directed graph (networkx)
 - ☒ Generate interactive visualization (pyvis)
 - ☒ Export list of isolated nodes (no connections to other articles in collection)
 - ☒ Export list of most-cited nodes within the collection (hubs)
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Phase 1.5: Citation Graph Improvements (DONE)

Objective: Fix known issues and add usability features to the citation graph.

Bug fixes

- ☒ Investigate missing edge: "The estimation of multivariate normal density functions using incomplete data" — root cause was multi-version papers on OpenAlex (e.g., Rubin 1975 ETS vs Rubin 1976 Biometrika). Fixed by reworking `find_canonical_version()` into `find_all_versions()` to collect all version IDs instead of just one
- ☒ Check if similar matching failures affect other papers — fix applies globally to all papers in the collection
- ☒ Fix isolated preprints (e.g., Mahmud 2025 ImpuGAN): arXiv versions often have 0 references on OpenAlex. Now fetches references from published alternate versions when the original has none
- ☒ Fix .bib parser compatibility with Zotero re-exports: handle `date` field (not just `year`) and accept non-standard entry types like `@report` (Dempster 1977, Murray 1979, etc. were being silently skipped)

New features

- ☒ Search bar: autocomplete dropdown that filters and highlights matching nodes, click to zoom
- ☒ Timeline view: static matplotlib timeline (papers on year axis with citation arrows)
- ☒ View switcher: top-right menu toggles between "Interactive View" (graph) and "Timeline View" (timeline) — replaces the old overlay approach
- ☒ Click-to-highlight: clicking a node dims all unconnected nodes/edges, highlights the selected node and its neighbors. Click empty space to restore.
- ☒ Color legend: always-visible legend (bottom-left) showing era colors in both views
- ☒ Improved node spacing: stronger repulsion and longer springs to reduce overlap

Phase 1.7: Address Peer Review Gaps (CURRENT — prioritized over 1.6)

Objective: Strengthen the collection based on NotebookLM peer review feedback and professor meeting (2026-09-02). Focus on gaps that fit our scope.

Key decision (2026-09-03): Phase 1.7 is now prioritized *before* finishing Phase 1.6 (extraction matrix). Reason: the extraction matrix needs columns like "Assumed missingness mechanism" and "Relevance to taxonomy." Without understanding the MNAR landscape first, those columns can't be filled meaningfully.

Professor meeting feedback (2026-09-02):

- Professors confirmed the MNAR gap is the highest priority
- The central problem: modern DL imputation methods (BRITS, SAITS, CSDI) don't state or verify the theoretical conditions (missingness mechanism) under which they are valid — the justification that links Rubin's framework to the method is missing
- Fernando noted that Sun, Qin & Huang (2018) — "Missing Information Principle" — is the only paper in the collection that touches this. We traced its citations via OpenAlex but found they are survival analysis focused, not useful for the imputation bridge.
- Valeria finds it difficult to understand the types of MNAR — this is a reading priority

Scope decisions (2026-08-31): The peer review identified 7 areas of concern. We evaluated each against our review's focus (imputation methods for multivariate time series in air quality) and made the following calls. These decisions can be revisited if a journal reviewer or Facundo disagrees:

- **MNAR mechanisms** → **IN SCOPE**. Sensor saturation at SIMA is textbook MNAR. Our review should acknowledge this and cover sensitivity analysis approaches. Without this, a reviewer could argue our applied framing is naive.
- **Imputation uncertainty propagation** → **IN SCOPE**. The tension between Rubin's proper inference and deep learning's RMSE optimization is the central narrative thread of our review (from Rubin to SAITS). We need papers that bridge this.
- **Burst missingness / long gaps** → **IN SCOPE**. Multi-day sensor outages are a real SIMA problem. Relevant to our applied context.
- **Bidirectional info leakage** → **BRIEF MENTION**. Valid point about real-time vs. retrospective, but it's an operational concern, not a methodological gap. One paragraph in discussion.
- **Extreme value / regulatory peaks** → **BRIEF FUTURE WORK**. Important for SIMA's regulatory context but it's a specialized topic beyond our review's arc.
- **Causal inference (m-DAGs, TMLE)** → **OUT OF SCOPE**. Our review is about imputation methods, not causal epidemiology. Adding this would change the paper's focus entirely.
- **Spatial methods (kriging, INLA, GP)** → **BRIEF FUTURE WORK**. Our narrative follows a temporal arc (statistics → ML → deep learning). Geostatistics is a parallel branch, not part of that evolution.

Gap 1: MNAR mechanisms + Rubin-to-DL bridge (high priority) — IN PROGRESS

Approach: Used NotebookLM deep search (free, saves Claude tokens) instead of OpenAlex keyword search (too noisy for conceptual gaps). Results in [Notebooklm/research-report-bridging-missing-data.md](#).

NotebookLM found 11 candidate papers across 4 themes:

1. MNAR identification & sensitivity analysis (NARFCS, tipping-point analysis)
2. Theoretical validity of DL under different mechanisms (FragmGAN proving GAIN assumes MCAR, VAE conditions under MAR)
3. Deep generative models designed for MNAR (MIWAE, not-MIWAE, PSMVAE, GNR)
4. RMSE vs. inferential validity tension (van Buuren "imputation is not prediction", Beyond Accuracy benchmark)

Important caveat: These 11 papers are mostly about tabular data, not time series. The bridge between MNAR theory and time series DL (BRITS, SAITS) specifically is still a gap — will need to be addressed in writing.

Known gap in the 11 candidates: All are post-2018. Missing the older foundational work (1990s-2010s) on pattern-mixture models, selection models, and early MNAR theory that the newer papers build on. Plan: after reviewing the 11 candidates, run a second NotebookLM deep search targeting pre-2018 foundations.

- ☒ NotebookLM deep search #1: modern bridge papers (2018-2025) — 11 candidates found
- ☒ Review and triage the 11 candidates — 6 accepted, 5 skipped. Added to Zotero and .bib.
 - ADD: #1 NARFCS (Tompsett 2018), #3 FragmaGAN (Fang 2023), #5 MIWAE (Mattei 2019), #6 not-MIWAE (Ipsen 2021), #8 PSMVAE (Ghalebikesabi 2021), #10 Beyond Accuracy (2025)
 - SKIP: #2 Tipping Point, #4 Nazábal VAE, #7 GNR, #9 IVGAE, #11 van Buuren

- ☒ NotebookLM deep search #2: pre-2018 foundational MNAR/pattern-mixture/selection model theory — results in [Notebooklm/classical-theoretical-foundations.md](#). Report covers 4 areas: selection vs. pattern-mixture models, MNAR taxonomy, sensitivity analysis origins, and ignorability foundations.
- ☒ Triage second batch — 5 accepted. Added to Zotero and .bib.
 - ADD: Little (1993) pattern-mixture models, Diggle & Kenward (1994) informative dropout, Little (1995) dropout taxonomy, Scharfstein, Rotnitzky & Robins (1999) semiparametric sensitivity, Carpenter, Kenward & White (2007) delta-adjustment method
 - Already in collection: Rubin (1976), Little & Rubin (1987/2002), Dempster et al. (1977), Heckman (1976, 1979)
- ☒ Re-run citation graph to verify the bridge connects
- ☒ Manual reference tracing: searched "neural network for imputation rubin little" and found additional relevant papers:
 - Gupta & Lam (1996) "Estimating Missing Values Using Neural Networks" — one of the earliest papers using NNs for imputation, historical marker for when ML entered the imputation space without Rubin's theoretical grounding
 - Rey del Castillo (2012, Eurostat WP.37) "Use of Machine Learning Methods to Impute Categorical Data" — explicitly quotes Rubin (1996) on why RMSE/hit-rate is the wrong evaluation criterion for imputation, then shows ML classifiers beating MI on exactly that metric (87% vs 66%). Perfect illustration of the inference-vs-accuracy tension.
 - Richard & Lippmann (1991) "Neural Network Classifiers Estimate Bayesian a posteriori Probabilities" (found via Hanson 1992 references) — proves NN classifiers approximate Bayesian posteriors. The equivalent theorem for NN *imputation* (proving convergence to Rubin-valid posterior predictive distributions) appears not to exist yet — this is a key observation for the discussion section.

Gap 2: Imputation uncertainty propagation (high priority)

- ☐ Search for papers on Rubin's pooling rules applied to deep learning, conformal prediction intervals for time series imputation
- ☐ Triage and add to collection
- Note: partially covered by Gap 1 candidates — van Buuren (2018) and "Beyond Accuracy" (2025) address this directly

Gap 3: Burst missingness / long consecutive gaps (medium priority)

- ☐ Search for papers specifically addressing multi-day sensor outages or long gap imputation
- ☐ Triage and add to collection

Discussion section additions (for Phase 4 writing)

- ☐ Point reconstruction vs. inferential validity — the field traded rigor (Rubin) for accuracy (SAITS). Core narrative thread.
- ☐ Bidirectional info leakage — retrospective reconstruction vs. real-time alerting. Brief paragraph.
- ☐ Extreme value / regulatory peaks — RMSE-optimized models underpredict toxic peaks. Brief future work.
- ☐ Spatiotemporal methods — mention as a future direction, not part of our arc.

Phase 1.6: Collection Analysis (PAUSED — waiting on 1.7)

Objective: Use the citation graph to understand the collection's strengths, gaps, and guide what to read or add next.

Tasks

- ☒ Year distribution analysis: identified thin spots in 1980-1999 (practical ML era), 2000-2014 (ML bridge — only 6 relevant papers), and 2015-2018 (early deep learning imputation like GAIN/BRITS)
- ☒ OpenAlex search tool ([search_papers.py](#)): searches by topic + year range, filters out papers already in collection, sorted by citations. Logs all searches to [search_log.csv](#) for methodology traceability
- ☒ Candidate paper triage: used NotebookLM to compare candidate papers (from search tool) against current collection. 3-step process: (1) baseline analysis of current collection's coverage and blind spots, (2) novelty filter to find candidates that fill gaps, (3) generate ADD/SKIP verdicts with rationale. Results saved to [Notebooklm/candidates_veredict.csv](#). Updated [.bib](#) with accepted papers.
- ☐ Extraction matrix template: build a structured CSV/table matching Phase 2 columns, ready to fill in as the team reads papers (blocked on finishing 1.7 — need MNAR understanding first)

Phase 2: Literature Extraction Matrix

Objective: Create a structured table to enable comparative writing (arguments, not summaries).

Columns (from Facundo's recommendation):

Column	Description
Reference	Author, year
Application domain	Air quality, health, traffic, energy, etc.
Data type	Univariate / Multivariate time series
Number of variables	Count
Temporal length	Duration / frequency of data
Assumed missingness mechanism	MCAR, MAR, MNAR
Evaluated missing pattern	Random, block/gap, mixed
Method	Name of imputation method
Output type	Point or probabilistic
Training loss	Loss function used
Metrics	RMSE, MAE, R2, etc.
Baselines	What they compared against
Advantages	Key strengths
Limitations	Key weaknesses

Column	Description
Code/data available	Yes/No + link
Relevance to taxonomy	Where it fits in the review's narrative

Status: Not started. Depends on Phase 1 to filter relevant articles.

Phase 3: Review Methodology Section

Objective: Write section 2 of the article (methodology of the review itself).

Subsections (from Fernando's outline):

- 2.1 Type of review (narrative, interpretive synthesis)
- 2.2 Sources consulted (backward/forward search from seminal articles)
- 2.3 Search strings (table by topic: fundamentals, time series, deep learning, generative, applications)
- 2.4 Inclusion criteria
- 2.5 Exclusion criteria
- 2.6 Synthesis strategy (organize by inferential foundations, method family, output type, missingness mechanism, data structure, computational cost, application domain)

Status: Not started. Fernando proposed the structure on Aug 23.

Phase 4: Article Writing

Narrative arc: Classical statistical foundations (Rubin, EM, MI) --> Traditional methods (interpolation, regression, KNN, Kalman) --> Machine learning (SOM, MLP, RF) --> Deep generative models (GANs, VAEs, diffusion, transformers/SAITS/CSDI) --> Application to air quality --> Limitations & open challenges

Status: Not started. Depends on Phases 2 and 3.

Notes from Facundo (Aug 24)

- SIMA data available since 1997; can be standardized
- Must review computational capacity to choose appropriate methods
- Climate in 1997 was very different from today — check validity, train multiple models, EDA, prior bias
- Don't discard old data — behavior/shape should be similar
- Prediction model: next-day forecasts, model confidence
- SIMA constraint: 15-minute publishing window (7 min data collection + 7 min calculation, not parallel)
- Machine limitations at SIMA

Potentially Off-Topic Articles (to verify with graph)

These entries in the .bib don't seem to match the review's scope:

- Lee 2005 — mobile ad hoc network routing
- Sarrouy 2009 — intrusion detection

- Zaki 2019 — 3D spatial modeling in MADS
 - Espejo Chahuara 2024 — municipal HR management (abstract doesn't match title)
 - Dong 2024 (SpecAR-Net) — abstract about multiwinner elections, not time series
 - Hamidi 2024 — customer behavior segmentation
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Tools & Setup

- **Reference manager:** Zotero (migrated from Mendeley on Aug 24)
- **Language:** Python
- **APIs:** Semantic Scholar, OpenAlex (free, no key needed for basic use)
- **This repo:** Tooling only (not the manuscript)